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Natural Language Processing: Morphological processing

H02B1A



Miryam de Lhoneux

Department of Computer Science
KU Leuven

Outline

Introduction

Morphology

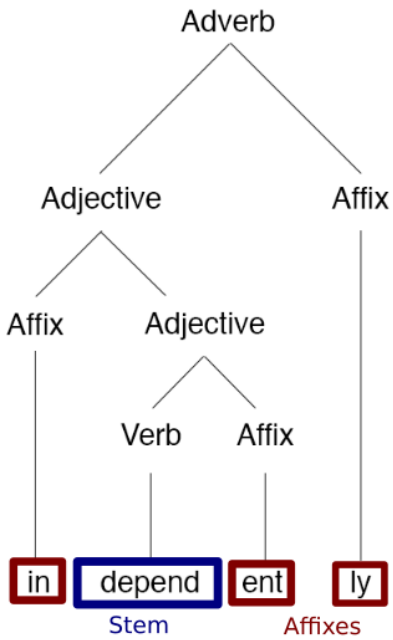
Processing morphology: tasks

Rule-based morphological processing

Data-driven morphological processing

Morphologically-informed subword tokenization

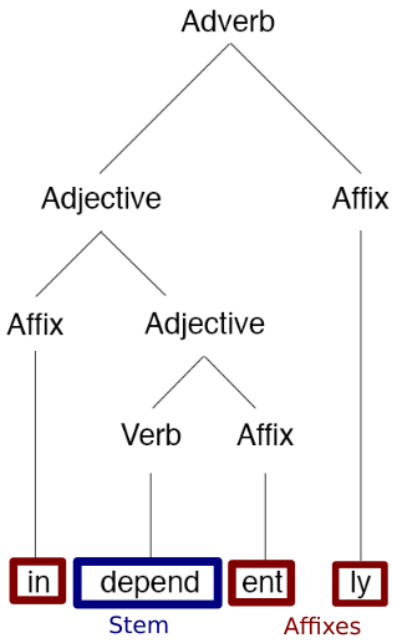
Summary



Morphological analysis of the word *independently* 🔗

Morphology

The study of the internal structure of words.



Morphological analysis of the word *independently* 🔗

Morphemes

The smallest unit of meaning.

Stem: *(or lemma)* supplies the *main* meaning

Affix: adds *additional* meaning

Why is it useful?

Example morphological analysis

	The	dogs	chase	the	cat	and	they	fall
Number		Plur			Sing		Plur	

- Informs natural language understanding e.g. coreference resolution

  Useful in downstream tasks e.g. question answering:

  Q Who is falling?

  A The dogs

Why is it useful?

Example morphological analysis

	The	dogs	chase	the	cat	and	they	fall
Number		Plur			Sing		Plur	

- Informs natural language understanding e.g. coreference resolution
- Helps with natural language generation e.g. machine translation.

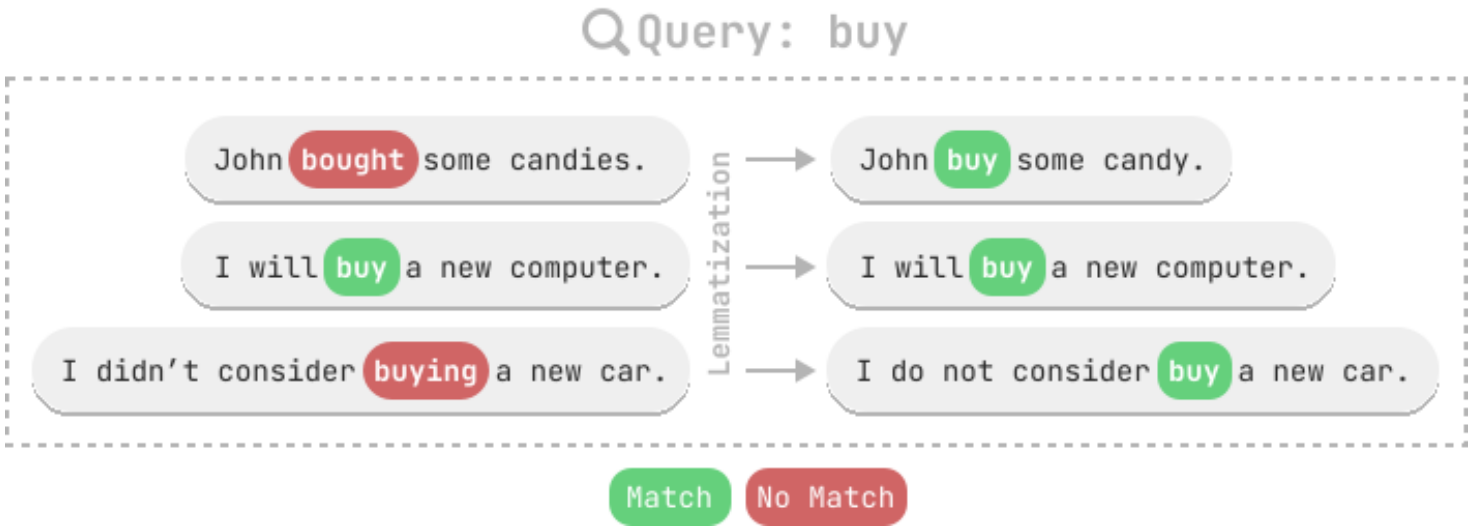
  Source (English): The dogs
  Target (French): Les chiens

Why is it useful?

Indirectly

Can help preprocess or normalize text (see also lecture 2).

E.g. lemmatization for information retrieval

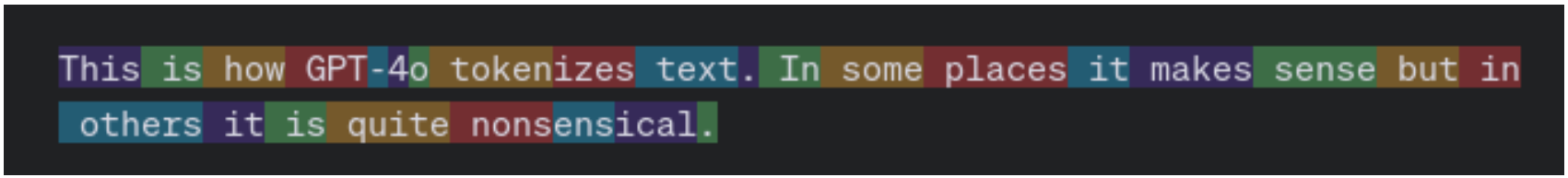


Example of the use of lemmatization for a search engine 🔗

Why is it useful?

Indirectly

Can inform subword tokenization



Example tokenization result by GPT-4o. Try it out yourself: 🔗

frequency-based: `nons-ens-ical`

morphologically informed: `non-sens-ical`

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Morphemes

The smallest unit of meaning.

Stem: supplies the *main* meaning

Also called lemma, base form, root, lexeme

Affix: adds *additional* meaning

Morphemes

Bound: Cannot appear as a word by itself

`s` (dog-`s`), `ly` (quick-`ly`), `ed` (walk-`ed`)

Free: Can appear as a word by itself, may combine with other morphemes

`of`, `the`, `dog` (`dog`-s), `walk` (`walk`-ed)

Inflectional morphemes

- Examples:

`dog`	(NOUN)	+ `s`	= `dogs`	(NOUN)
`eat`	(VERB)	+ `s`	= `eats`	(VERB)
`eat`	(VERB)	+ `ing`	= `eating`	(VERB)

- Bound morphemes used to make grammatical distinctions
- Change the form of the word but not its POS tag or key meaning

Derivational morphemes

- Examples:

`appoint`	(VERB)	+ `ee`	= `appointee`	(NOUN)
`help`	(VERB)	+ `less`	= `helpless`	(ADJ)
`strength`	(NOUN)	+ `en`	= `strengthen`	(VERB)

- Bound morphemes used to create new words
- Change the form, POS tag and meaning of the word

Concatenative morphology

Stem + affix(es).

Affixes

- Prefixes: `disestablish`
- Suffixes: `establishment`

Can be combined: `disestablishment`

Non-concatenative morphology

Infixes: e.g., Tagalog:
`hingi` (borrow) → `humingi` (borrower)

Circumfixes e.g., German:
`sagen` (say) → `gesagt` (said)

Reduplication e.g., Motu:
`mahuta` (to sleep) → `mahutamahuta` (to sleep constantly)

Morphological typology

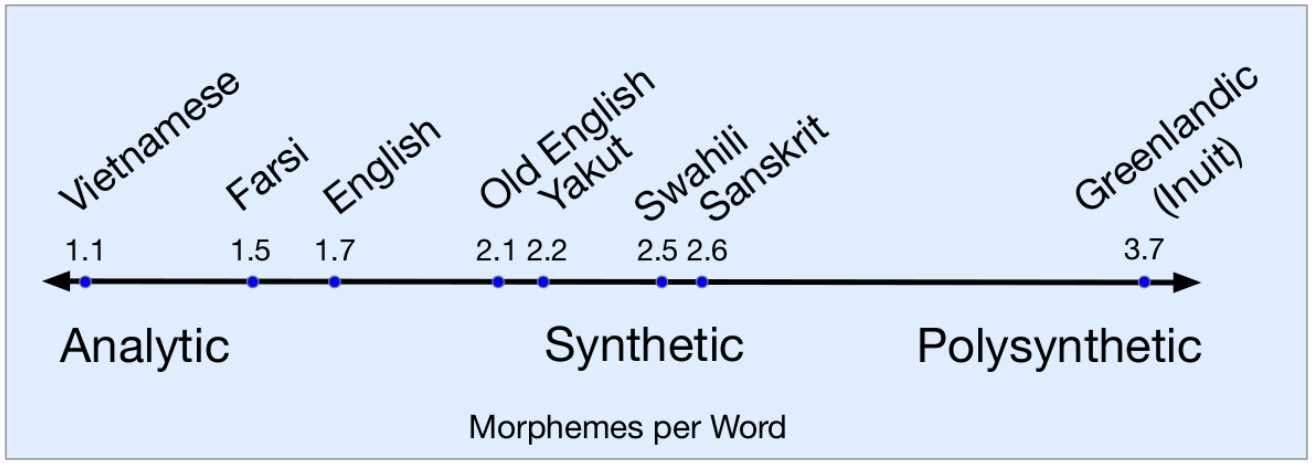
Languages vary in how they use morphology

English:	We learned these new words					
Chinese:	Wǒmen I.PL.AN	xué learn	le .PAST	zhè this	xi .PL	shēngcí new word
Russian:	My We.NOM	vyučili learn.PAST.PL	eti this.ACC.PL	novyje new.ACC.PL	slova word.ACC.PL	

- Russian (synthetic) has more morphemes per word than Chinese (isolating)
- English is in between, it's a continuum

Examples borrowed from Kat Vyloma 🔗

Morphological typology



Languages on a continuum of morphemes per word, estimated by Greenberg (1960), figure by Jurafsky and Martin (2025)

Morphological typology

West Greenlandic (polysynthetic)

Nannu-n-niuti-kkuminar-tu-rujussu-u-vuq
polar.bear-catch-instrument.for.achieving-something.good.for-PART-big-be-3SG.INDIC
It (a dog) is really good for catching polar bears with.

Example from Fortescue (2017)

Morphological typology

Agglutinative: Morphemes have one meaning (e.g. Turkish)

Fusional: Morphemes may have multiple meanings (e.g. Russian)

This is also a continuum

Morphological typology

- Agglutinative and fusional languages are more morphologically complex than isolating languages
- Large consensus that morphologically complex languages are underserved by current NLP models
- Ongoing research on:
  - quantifying and explaining the difficulty, e.g. Park et al. (2021); Poelman et al. (2025)
  - addressing the inequality, e.g. Ataman et al. (2020)

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Morphological analysis

Tasks

Input: `puppies`

- Lemmatization: `puppies` → `puppy`
- Segmentation: `puppies` → `puppy|s`
- Tagging: detailed morphological summary

  Lemma: puppy
  POS: NOUN
  Number: plural

Analysis *in isolation*

Input: `saw`

1. Lemma: see
   POS: VERB
   Number: singular
   Tense: past
2. Lemma: saw
   POS: VERB
   Number: singular
   Tense: present
3. Lemma: saw
   POS: NOUN
   Number: singular

Analysis *in context*

Input: `cooler`

- Context: `The cooler was full of orange juice`

  Lemma: cooler
  POS: NOUN
  Number: singular

- Context: `The daisy prefers cooler weather`

  Lemma: cool
  POS: ADJ
  Degree: comparative

Morphological generation

Inflection generation

Given a lemma and a morphological specification (inflection), generate the appropriate inflected form.

English:	permaban	V;V.PTCP;PST	permabanned
German:	Stiefvater	N;ACC;PL	Stiefväter
Hungarian:	homeosztázis	N;DAT;PL	homeosztázisoknak

Morphological generation

Paradigm generation

Given a lemma, generate the full inflectional paradigm.

`schleichen`:

schleiche	V;1;SG;PRS
schleichst	V;2;SG;PRS
schleicht	V;3;SG;PRS
geschlichen	V;V.PTCP;PST
...	

Morphological generation

Generation in context

Given a lemma, generate the correct inflection form using sentential context.

Input: `The dog are barking`

Output: `dogs`

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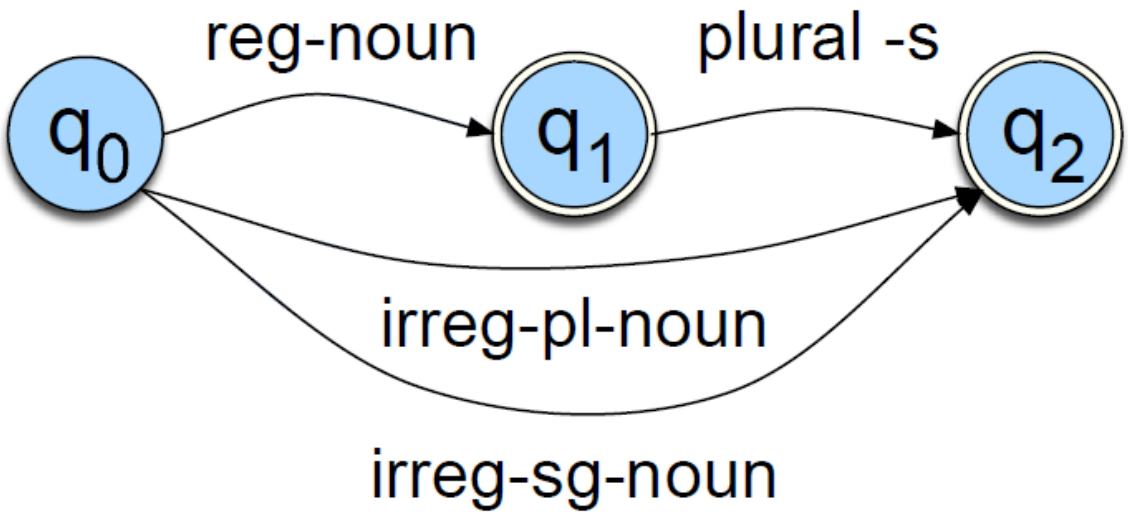
Summary

English noun inflection

	Singular	Plural
Regular	cat	cats
	thrush	thrushes
Irregular	mouse	mice
	ox	oxen

Finite-state automaton (FSA)

An abstract machine used to recognize patterns in input sequences.
(Used also to implement regular expressions, see lecture 2)

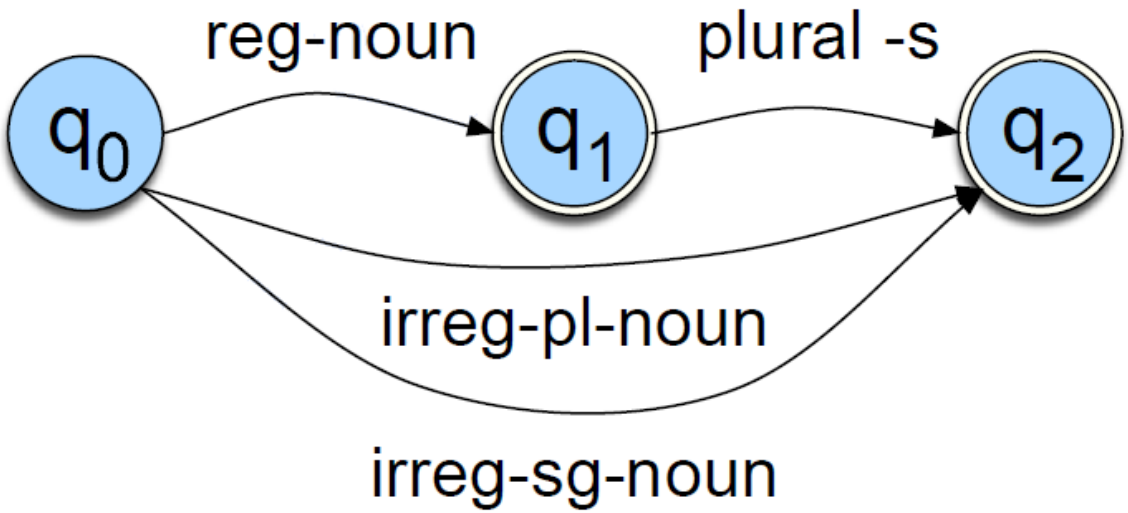


A FSA for English nominal inflection (Jurafsky and Martin, 2009)

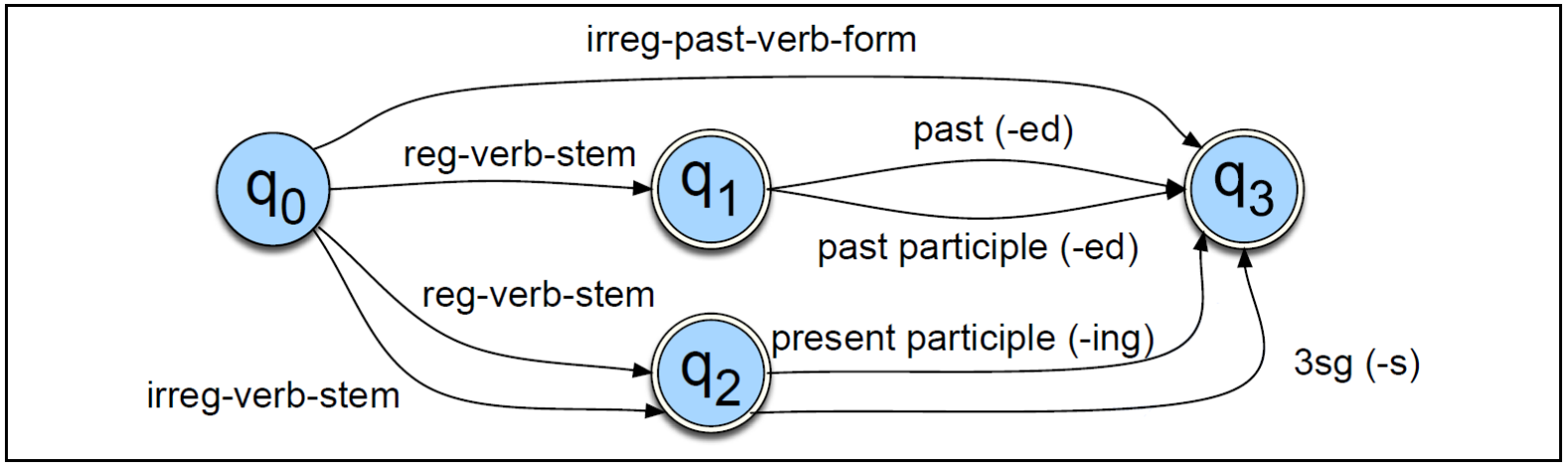
Needs

- A list of stems (*reg-noun*)
- A list of singular and plural irregular nouns

$\rightarrow$ A lexicon



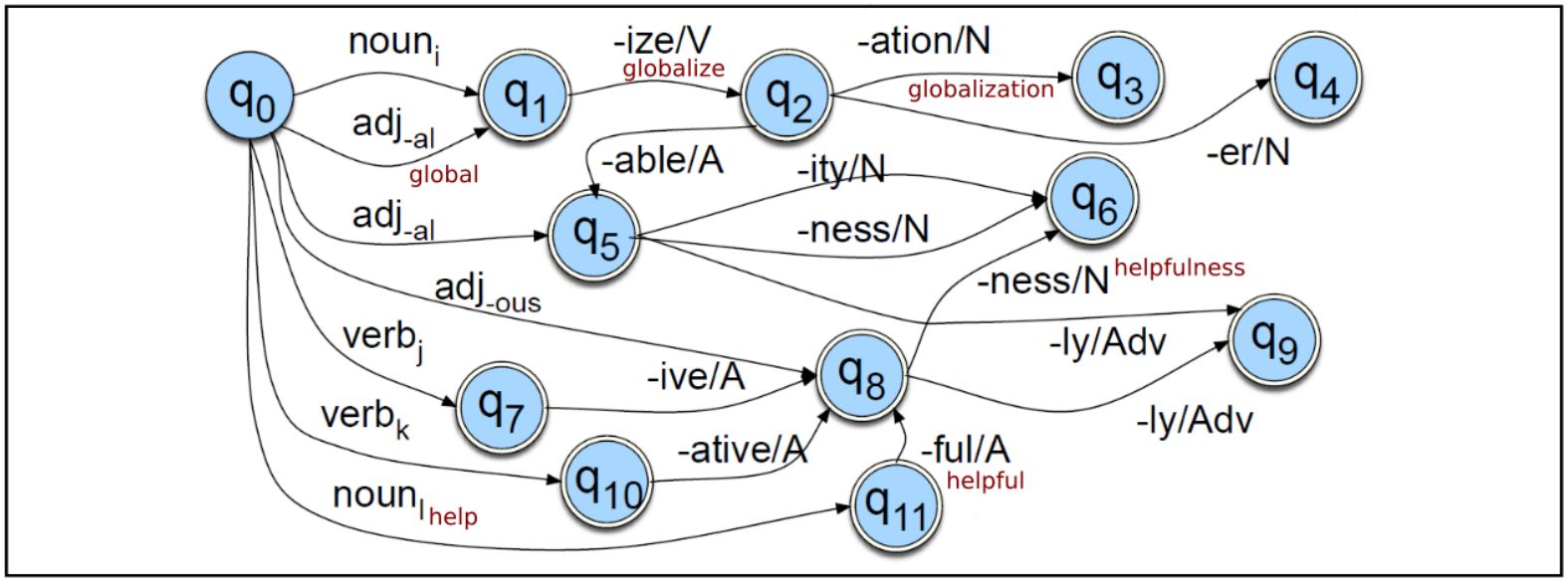
A FSA for English nominal inflection (Jurafsky and Martin, 2009)



A FSA for English verbal inflection (Jurafsky and Martin, 2009)

reg-verb-stem:	walk	fry	talk
irreg-verb-stem:	catch	speak	sing
irreg-past-verb-form:	caught	spoke	sang

- walk + past = walked
- walk + present participle = walking
- walk + 3sg = walks



A FSA for English nominal inflection (Jurafsky and Martin, 2009)

Use of FSAs

- Word recognition
  - Helpfulness ✔
  - Helpfully ✔
  - Helpfulization ✘
- Generalized to Finite State Transducers (FSTs)
  - Morphological analysis
  - Morphological generation

Limitations of FSAs

- Require linguistic annotation which is costly
- Need to be updated when new words appear in the vocabulary
- Cannot handle reduplication where arbitrarily long sequences are copied

We won't see more details. See Chapter 3 of Jurafsky and Martin (2009) if you are interested.

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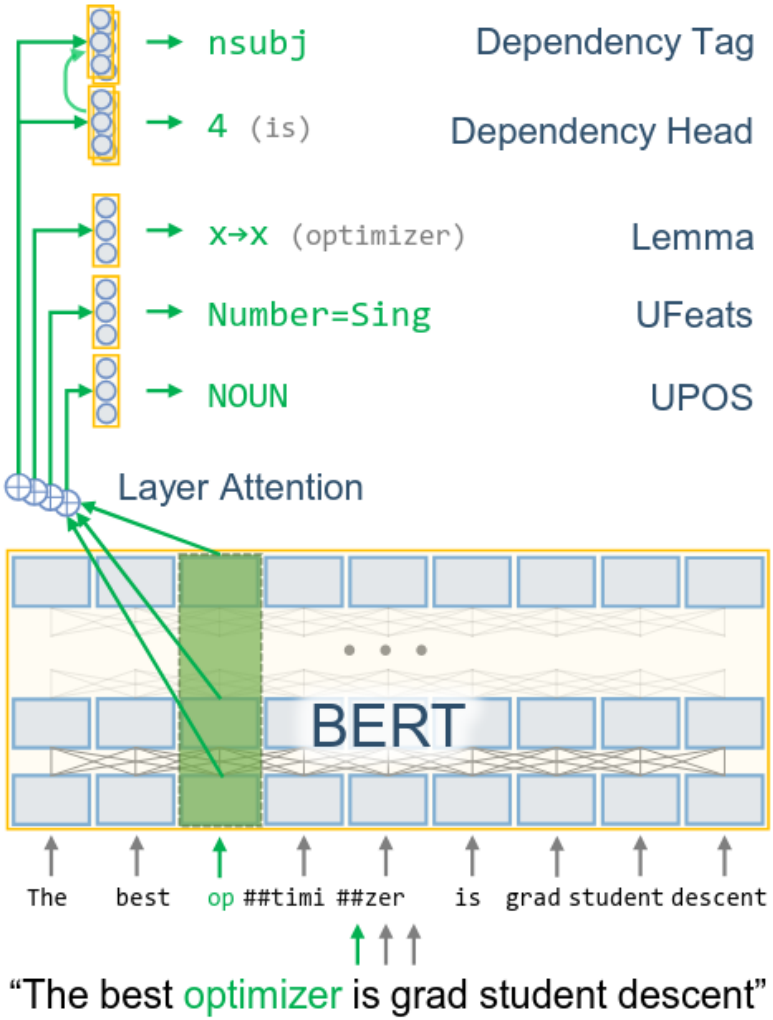
Morphologically-informed subword tokenization

Summary

Morphological analysis

- Most tasks in morphological analysis can be framed as word-level tagging tasks
- Tagging models can be applied directly (see lecture 6)

	Lemma	Morphology
Le	le	Definite=Def|Gender=Masc|Number=Sing
chat	chat	Gender=Masc|Number=Sing
chasse	chasser	Mood=Ind|Number=Sing|Person=3|Tense=Pres|VerbForm=Fin
les	le	Definite=Def|Gender=Masc|Number=Sing
chiens	chien	Gender=Masc|Number=Plur
.	.	_



Multitask architecture for joint POS, morphology, lemma tagging and dependency parsing (Kondratyuk and Straka, 2019).

Tagging in Kondratyuk and Straka (2019)

- Input representation: word represented using a transformer
- Passed to a softmax
- Argmax of softmax is selected as tag

Morphological analysis as tagging

- Tasks learned

  sequentially: lemmatizing → POS tagging → morphology

  jointly: Multi-task learning

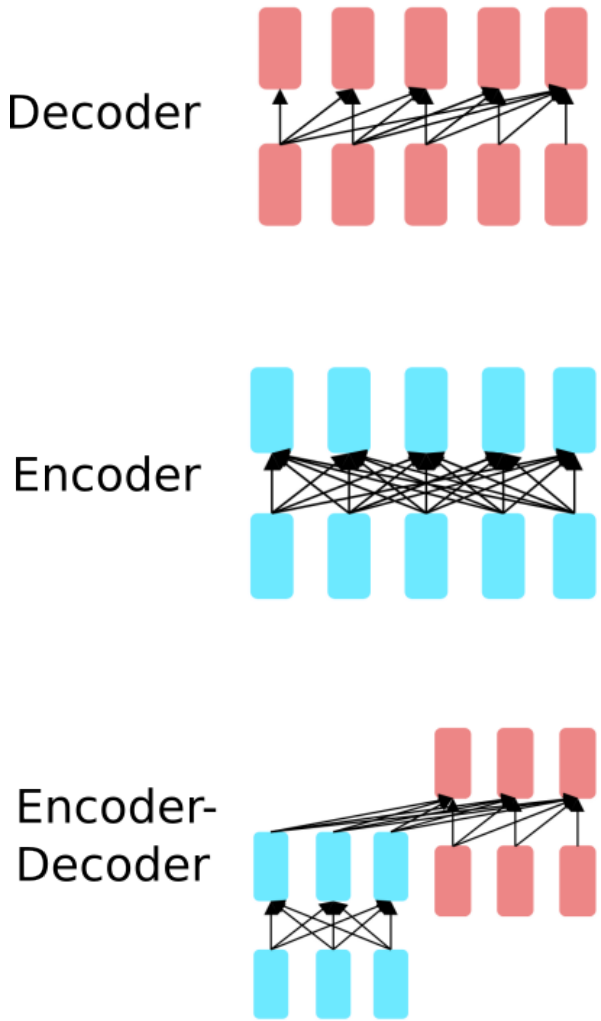
- Morphological tags:

  all-in-one: Tag: `Mood=Ind|Number=Sing|Person=3|Tense=Pres|VerbForm=Fin`

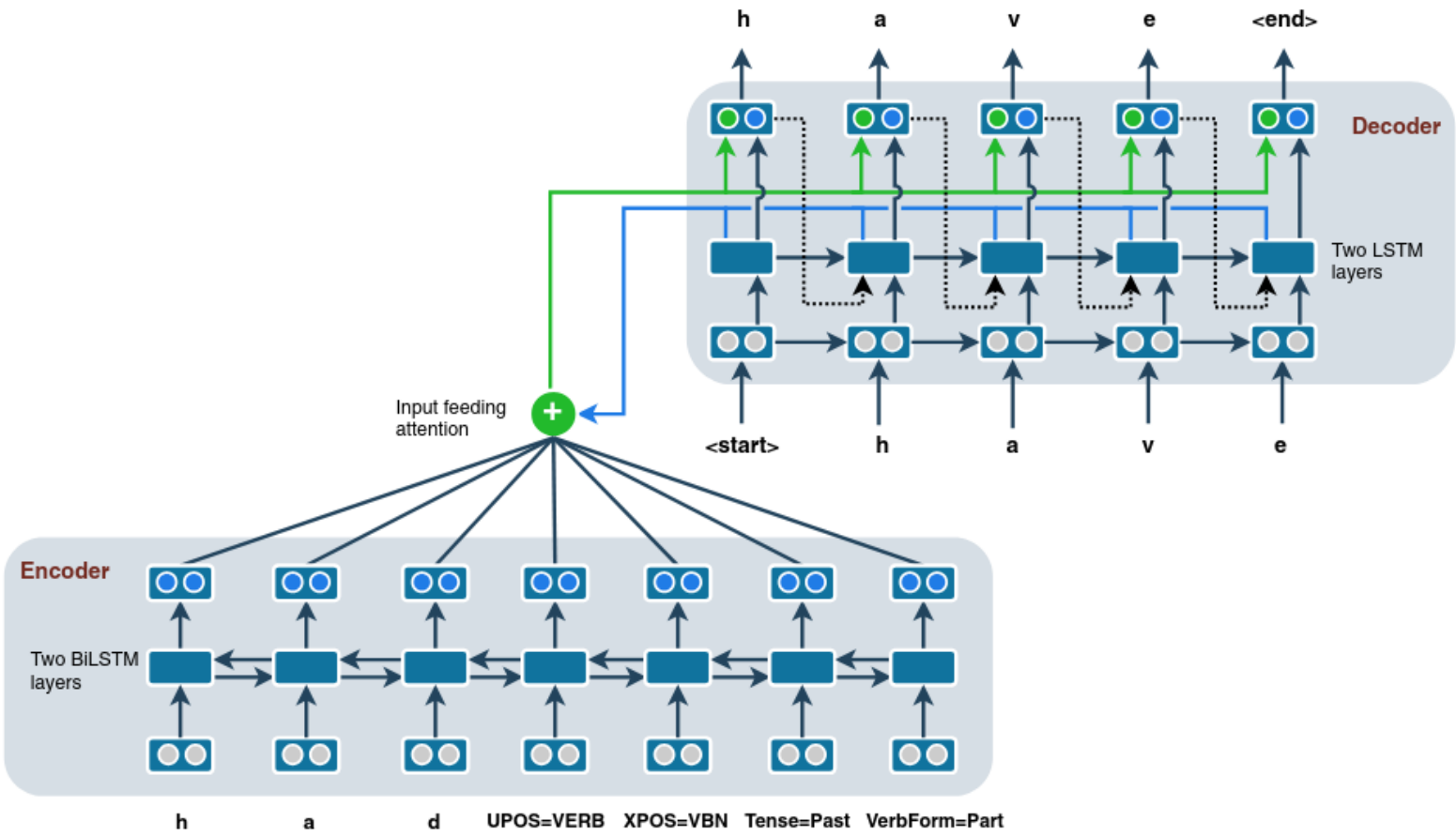
  split: Tag 1: `Mood=Ind`, Tag 2: `Number=Sing` etc.

In practice, joint learning of all-in-one category works best.

Instead of using an encoder architecture, some of these tasks can be modelled with an encoder-decoder architecture.



Visualization of the different NN architectures for sequence processing. See lecture 4.

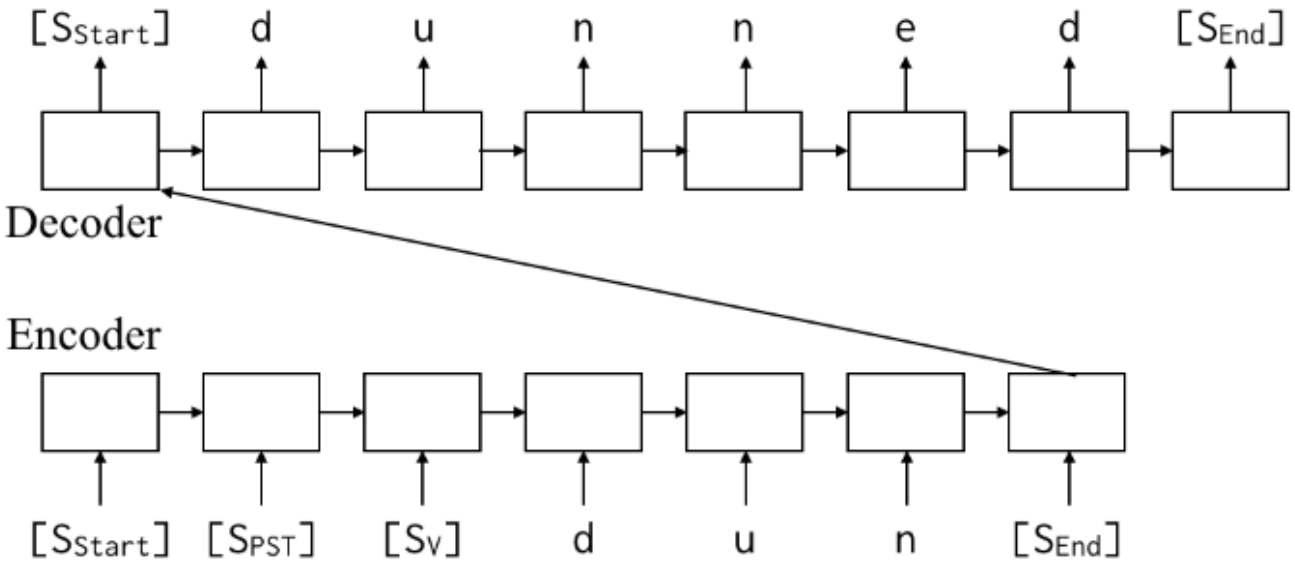


Encoder-decoder architecture for lemmatization (Kanerva et al., 2021).

The model exploits character-level information and reduces sparsity.

Morphological generation

For morphological generation tasks, the encoder-decoder (or sequence-to-sequence paradigm) can be applied.



Seq2seq model for inflection generation. The model produces the simple past form of the verb *dun* (Senuma and Aizawa, 2017).

Evaluation and SOTA

- Evaluation metric: depends on task but usually word-level accuracy
- SOTA for morphological analysis: lower than POS tagging with more variation ($\sim$50 to 95 if enough data)
- Even more language dependent
- Check out more on sigmorphon if you are interested

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Remember BPE? (Lecture 2)

Corpus

```
5  l o w _
2  l o w e s t _
6  n e w er_
3  w i d er_
2  n e w _
```

Vocabulary

_, d, e, i, l, n, o, r, s, t, w, er, er_

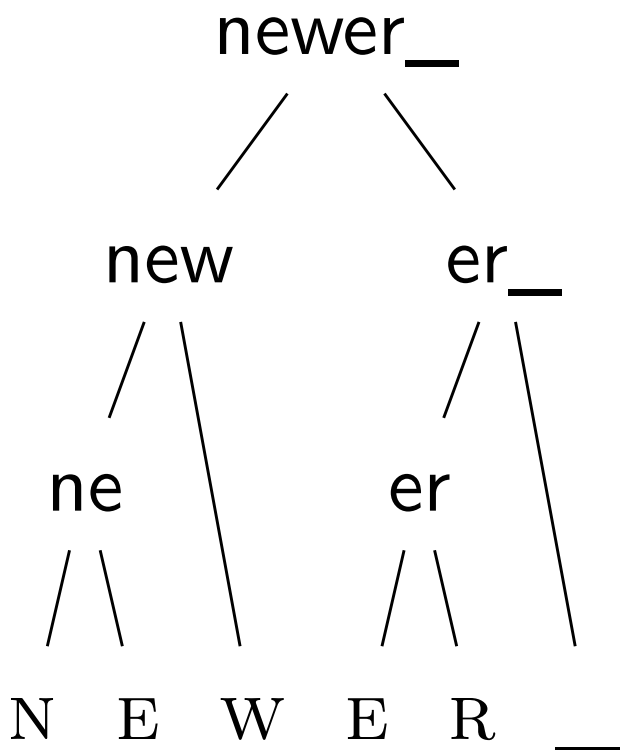
First merge

e r → er

Second merge

er _ → er_

Representing BPE merges as a tree



Merge rules

M1 `(e, r)`

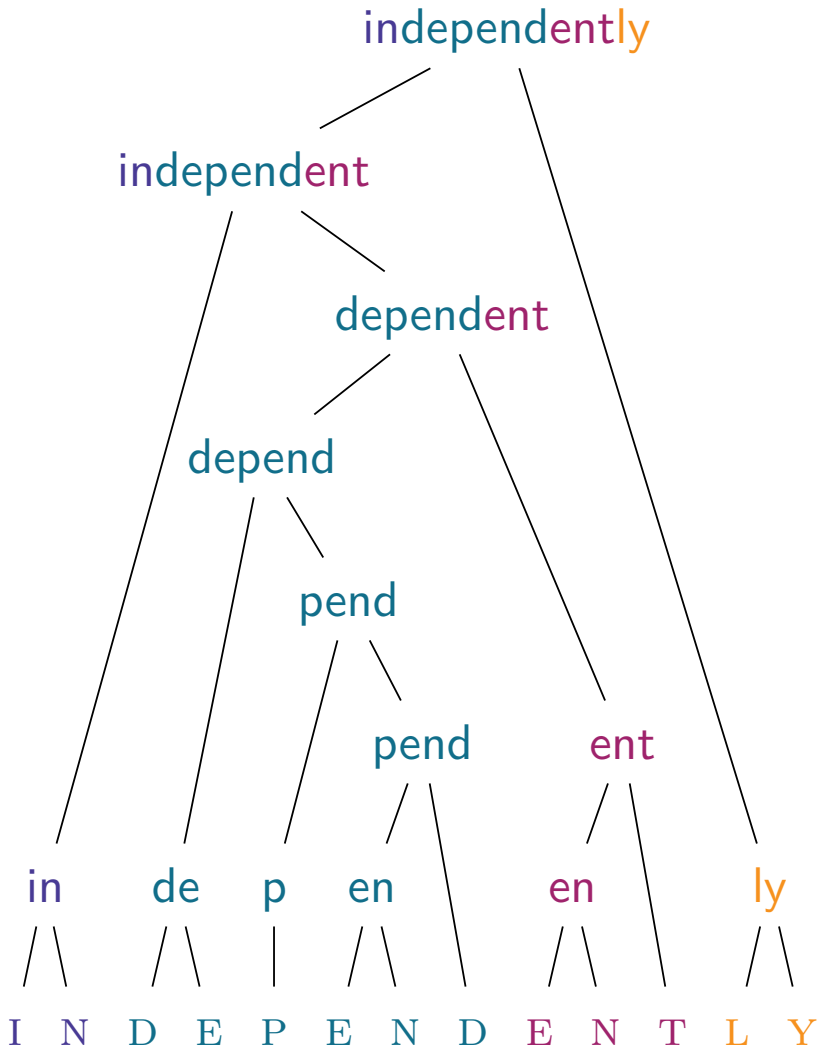
M2 `(er, _)`

M3 `(n, e)`

M4 `(ne, w)`

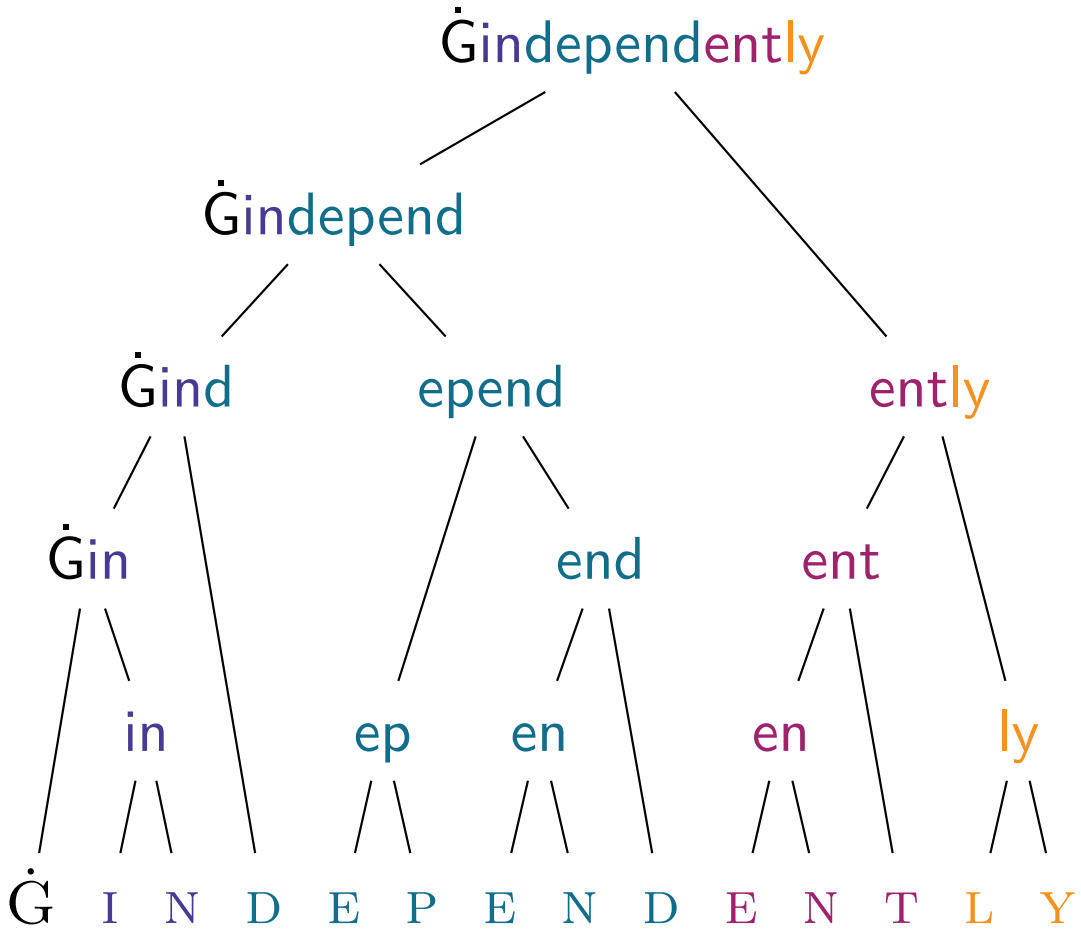
M7 `(new, er_)`

So the tree for *independently* should look roughly like this, right?



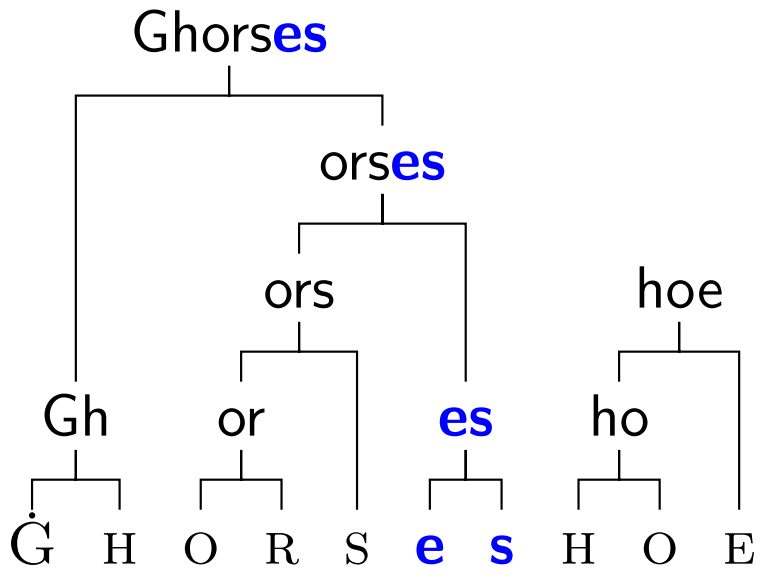
Imaginary BPE tree consistent with morphological boundaries.

Well...



RoBERTa BPE tree, generated with the TkTkT library by Thomas Bauwens.

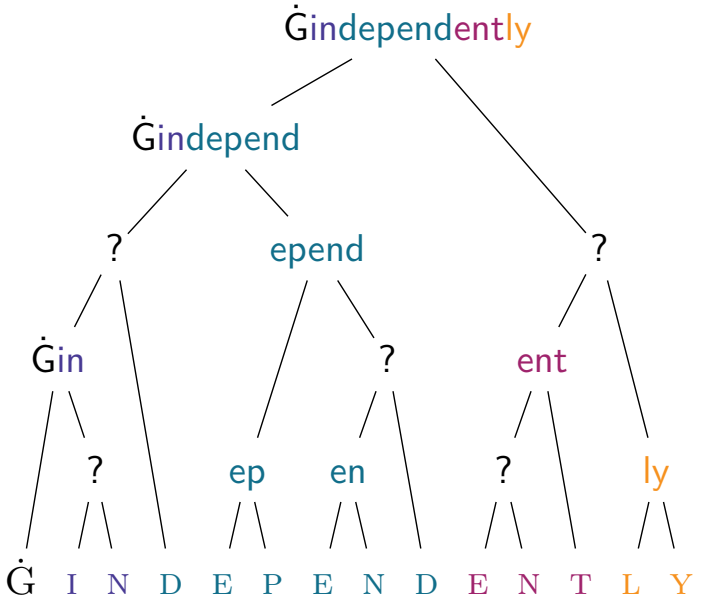
- You might say it does not matter if the end result is consistent with morphological boundaries
- But in some cases it is not



BPE tree of the word *horseshoe* by RoBERTa 🔗

- BPE segments text deterministically
- Some morphemes like ly may be in the vocabulary but underused in segmentation $\Rightarrow$ undertrained
- Limits compositional generalization:
  - initially
  - inherentently
  - dependable

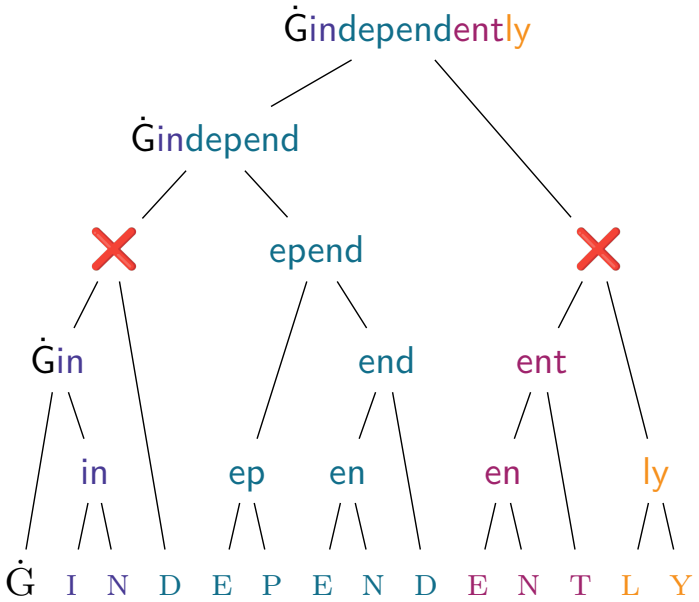
What can we do about it?



BPE-dropout (Provilkov et al., 2020)

- Randomly disable merges (temporarily)
- Can produce multiple segmentations of the same input
- Empirically improves downstream performance

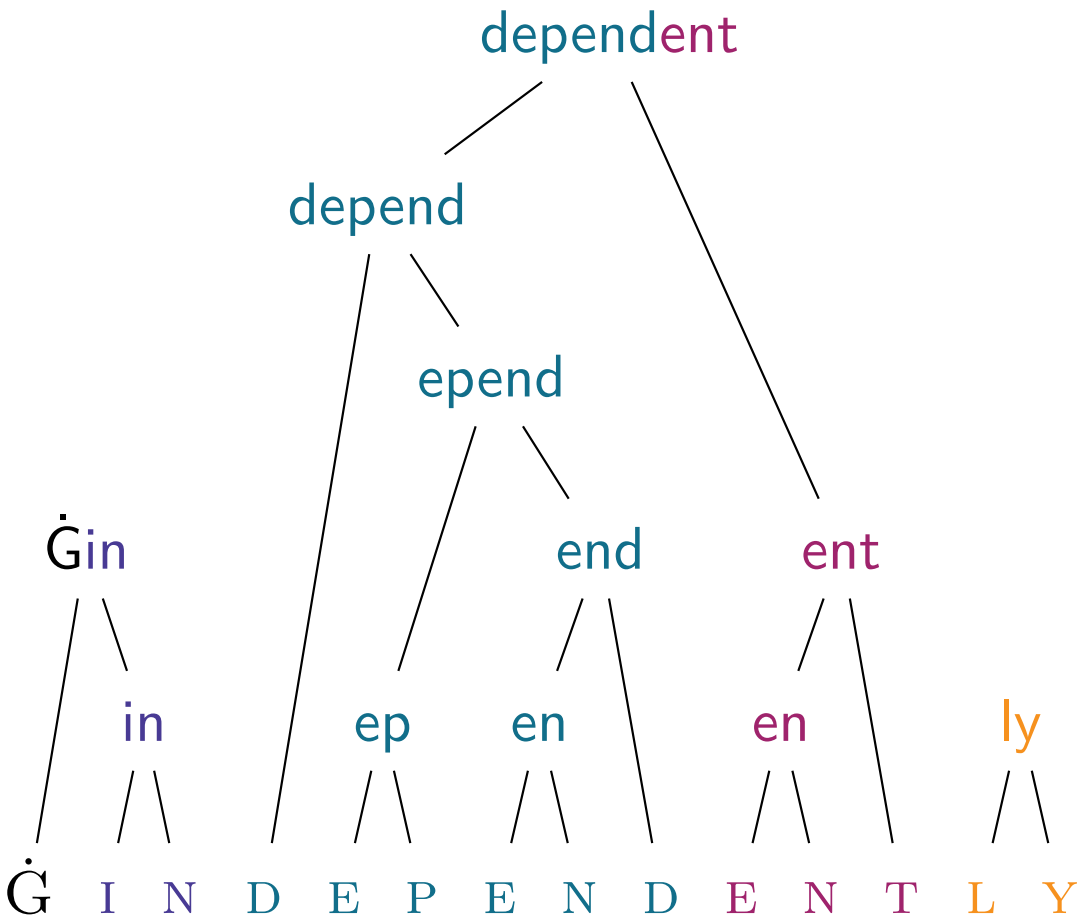
What can we do about it?



BPE-knockout (Bauwens and Delobelle, 2024)

- Disable *bad merges*
- Bad merges are determined based on a morphological lexicon
- Outperforms BPE-dropout downstream

BPE-knockout result for *independently*



BPE-knockout tree, generated with the TkTkT library by Thomas Bauwens.

BPE-knockout conclusion

- Morphological information can be used to inform subword tokenization
- This improves downstream performance
- Disadvantage: relies on handcrafted lexicon
- Still room for improvement

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- Morphology: the study of the internal structure of words
- Morphological processing tasks: tagging and generation
- FSAs: rule-based methods applied to these tasks
- Neural network based methods
- How to inform subword tokenization using morphology

Outlook

- Direct use of morphology is no longer relevant for English
- May still be relevant for languages with more complex morphology
- Models of morphological processing can still be improved for some languages
- Indirect use of morphology in subword tokenization holds promise **even for English**

Acknowledgments

The morphology slides took inspiration from Jimmy Lin 🔗.
The processing morphology tasks slides took inspiration from Kyle Gorman 🔗.

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